

COMMENT PROMPT

Create a simple **Python Area Calculator program** using a `while True` loop and the `math` module.

The program should:

1. Display a menu repeatedly:
 - 1. Circle
 - 2. Square
 - 3. Rectangle
 - 4. Exit
2. Ask the user to enter their choice.
3. If the user chooses:
 - **Circle:** Ask for the radius and calculate the area using `math.pi * radius * radius`.
 - **Square:** Ask for the side and calculate the area using `side * side`.
 - **Rectangle:** Ask for the length and breadth and calculate the area using `length * breadth`.
 - **Exit:** Display "Thank you!" and stop the program using `break`.
4. Display the calculated area as an integer using `int(area)`.
5. If the user enters an invalid choice, display "Invalid choice!".
6. After every calculation or invalid choice, display "Calculation completed." using a `for` loop with `range(1)`.
7. Keep the program beginner-friendly and use simple Python syntax

```
import math
```

```
while True:
```

```
    print("\n===== AREA CALCULATOR =====")
```

```
print("1. Circle")
print("2. Square")
print("3. Rectangle")
print("4. Exit")

choice = int(input("Enter your choice: "))

if choice == 1:
    radius = float(input("Enter radius: "))
    area = math.pi * radius * radius
    print("Area of Circle =", int(area))

elif choice == 2:
    side = float(input("Enter side: "))
    area = side * side
    print("Area of Square =", int(area))

elif choice == 3:
    length = float(input("Enter length: "))
    breadth = float(input("Enter breadth: "))
    area = length * breadth
    print("Area of Rectangle =", int(area))

elif choice == 4:
    print("Thank you!")
    break

else:
    print("Invalid choice!")

for i in range(1):
```

```
print("Calculation completed.")
```

OUTPUT

===== AREA CALCULATOR =====

1. Circle

2. Square

3. Rectangle

4. Exit

Enter your choice: 1

Enter radius: 5

Area of Circle = 78

Calculation completed.

===== AREA CALCULATOR =====

1. Circle

2. Square

3. Rectangle

4. Exit

Enter your choice: 2

Enter side: 4

Area of Square = 16

Calculation completed.

===== AREA CALCULATOR =====

1. Circle

2. Square

3. Rectangle

4. Exit

Enter your choice: 3

Enter length: 10

Enter breadth: 5

Area of Rectangle = 50

Calculation completed.

===== AREA CALCULATOR =====

1. Circle

2. Square

3. Rectangle

4. Exit

Enter your choice: 4

Thank you!